Structure agreement: BChemXtract 1.4.1 vs RDKit, Indigo, ChemScanner and Open Babel

Do the readers extract the same molecules from the same ChemDraw files? A per-structure comparison, error type by error type

Generated 2026-10-15, revised 2026-10-16 · 225 `.cdx` + 21 `.cdxml` files under `testset/corpus` (Open Babel: `.cdx` only) · reference BChemXtract 1.4.1 (`resolveRGroups = false`) vs RDKit 2026.09.1, Indigo 1.46.0, ChemScanner 0.1.3, Open Babel 3.1.1 · compared with `testset/compare_structures.py`

> **How to read this.** The corpus has no ground truth. BChemXtract is the reference every other reader is compared against, so these numbers measure *agreement with BChemXtract*, not correctness. When two readers disagree, either one can be wrong; where this report says which, it was checked against the drawing. All structures are compared through their SMILES, including BChemXtract's, whose SMILES and InChIKey do not always agree (see findings).

At a glance

EXACT AGREEMENT

65–84%

Paired structures identical to BChemXtract's; Indigo highest, Open Babel lowest

STEREO DISAGREEMENT

9–12%

Same skeleton, stereo differs; the largest disagreement for every reader

CONSENSUS AGAINST BCHEMXTRACT

10

InChIs where 2–4 readers agree on other stereo; 8 fixed in draft PR #227

VERIFIED MISREADING

59

Open Babel expands `t-Bu` as isobutyl

Pairing

STRUCTURES	RDKIT	INDIGO	CHEMSCANNER	OPEN BABEL
Paired by identical InChIKey	826	835	686	518
Paired by identical skeleton (InChIKey block 1)	98	101	102	70
Paired by fingerprint similarity (Tanimoto ≥ 0.5)	123	62	39	193
Compared (pairs minus references without InChI)	1027	976	815	764
Only in BChemXtract	363	412	583	629
Only in the other reader	893	540	732	708
Left out: RDKit hangs on the structure	5	11	10	8

Per file, both sides are deduplicated on InChIKey and paired greedily, best evidence first. Unpaired structures are counted but not compared: mostly label-only fragments and `*` pieces with no counterpart, plus Markush and S-group output that only one side produces. The left-out row counts BChemXtract's and the reader's structures together (5 BChemXtract structures from `bjnano_5_145` hang RDKit in every run).

Verdicts on the compared structures

VERDICT (SHARE OF COMPARED STRUCTURES)	RDKIT	INDIGO	CHEMSCANNER	OPEN BABEL
Exact: same molecule, same drawn form	78.2%	**84.3%**	81.5%	65.4%
Same InChI, different drawn form (tautomer, charge separation)	2.3%	1.8%	2.8%	2.4%
Wrong skeleton: other atoms or bonds	4.1%	2.7%	2.5%	11.3%
Stereo differs (same skeleton)	9.3%	10.0%	12.4%	9.0%
· conflict: a centre inverted or a double bond flipped	3.9%	2.7%	2.3%	1.6%
· reader has stereo BChemXtract leaves undefined	3.4%	0.6%	2.1%	1.3%
· reader leaves stereo undefined that BChemXtract assigns	1.9%	6.8%	8.0%	6.2%
Reader keeps `*` where BChemXtract has a full structure	5.8%	0.8%	0.7%	11.6%
Other layer / timeout / error	0.3%	0.3%	0.1%	0.3%

Stereo, counted per centre and per double bond

ROWS WITH THE SAME SKELETON	RDKIT	INDIGO	CHEMSCANNER	OPEN BABEL
Stereocentres agreeing (CIP R/S)	288	150	142	84
Stereocentres inverted	52	47	27	11
Stereocentres BChemXtract assigns, reader does not	69	195	228	139
Stereocentres the reader assigns, BChemXtract does not	70	11	14	4
Double bonds agreeing / flipped / missing / extra (E/Z)	5 / 1 / 0 / 0	9 / 0 / 20 / 6	7 / 0 / 0 / 8	0 / 1 / 5 / 7

Key findings

- **Most paired structures agree exactly.** 84.3% for Indigo, 81.5% for ChemScanner, 78.2% for RDKit and 65.4% for Open Babel. Where InChIKeys match, the drawn form differs in only 1.8–2.8% of pairs (tautomers and charge-separated forms that standard InChI deliberately merges).
- **Stereo is the largest disagreement, and the readers err in different directions.** Indigo, ChemScanner and Open Babel mostly leave centres undefined that BChemXtract assigns (195, 228 and 139 centres). RDKit assigns about as many centres that BChemXtract leaves open (70) as it leaves open itself (69), and inverts the most (52). Conflicts are spread over `vol18Test`, `vol19Test` and `bug`, not concentrated in `StereoInconsistency`; Indigo's include 5 sugar fixtures (`integrationTests/sugars`).
- **Where the other readers agree with each other against BChemXtract, BChemXtract is usually the outlier, but not always in its InChI.** 17 BChemXtract 1.4.1 SMILES have a stereo conflict with 2–4 readers that all give the identical alternative. Checked against BChemXtract's own InChIKey:
  - **7 are SMILES-only errors.** The InChI already matches the readers in 1.4.1; only the SMILES is inverted (`m28095160-1`, `-2`, `m47935605-i23`, `m25422595-i12`, and 3 spiro compounds in `bug/m43214419-1.cdx`). Current `main` writes the first four SMILES correctly; the three spiro SMILES are still inverted.
  - **10 are InChI errors**, in 1.4.1 and current `main` alike. 8 are in `vol18Test/m25422595-i8.cdx`: bold perspective bonds next to a real wedge were read as extra wedges. ChemDraw's stored labels agree with the readers; draft PR #227 fixes all 8. 1 is in `vol19Test/m45197327-i5.cdx`, a centre whose wedge and hash contradict each other in CDK's geometry. 1 is in `m43214419-1`, where ChemDraw itself leaves two of the three centres unspecified and the readers split.
- **BChemXtract's SMILES and InChIKey can describe different stereo.** On current `main`, for 138 of 1427 substances the InChIKey that CDK computes from BChemXtract's own SMILES differs from BChemXtract's InChIKey, always in the stereo block. The cause: the InChI is generated from a container that still carries the drawing's coordinates and wedge flags, so the InChI library derives the stereo from the drawing itself, while the SMILES is written from BChemXtract's stereo perception. Generated without coordinates, the InChI equals the SMILES in all 138.

    - 124 differ only in double-bond (E/Z) stereo, which the InChI library takes from the 2D geometry where BChemXtract's stereo perception assigns none. 78 of these are ring double bonds in the large polycycles of `bjnano_5_145`; in `m5445236-i5`, 36 of the 86 substances differ from another one only by this, so they are probably not distinct substances. 24 are in the porphyrin of `m31656373-1`, where BChemXtract leaves 4 of 8 double bonds unknown.
    - 14 differ at tetrahedral centres, among them the 3 spiro compounds of `m43214419-1`.
    - Where RDKit or Indigo extract the same structure, they match BChemXtract's InChI 19 times and the SMILES-side stereo never: the InChI is the more reliable output, and the SMILES loses stereo that the drawing defines.

  An earlier count of 52 (1.4.1) rested on RDKit reading BChemXtract's SMILES, which it does differently for 190 substances; the CDK round trip replaces it.
- **Open Babel misreads `t-Bu`.** 59 of its 86 wrong-skeleton pairs have the same formula as BChemXtract's structure: Open Babel expands the label `t-Bu` as isobutyl (`–CH2CH(CH3)2`) where BChemXtract gives tert-butyl. The drawings use `t-Bu` (79 times in `vol19Test/2-1.cdx`, 15 in `m25620245-5.cdx`), so Open Babel has the wrong molecule. These are valid InChIs for the wrong compound, which an InChI-coverage count cannot see.
- **RDKit and BChemXtract read one label differently.** In `StereoInconsistency/m29814417-9.cdx` BChemXtract has two pentafluorobenzoate esters where RDKit has two acetates (8 pairs, formula delta $C_{10}F_{10}$). The label is not stored as plain text, so which reading is right was not settled here.
- **Unexpanded labels, seen per structure.** For 89 Open Babel and 60 RDKit structures the reader keeps a `*` where BChemXtract produced the full molecule (8 for Indigo, 6 for ChemScanner). This is the per-structure view of the wildcard counts in the performance report.
- **Valid-looking output can be the wrong molecule.** Besides `t-Bu`, the performance report's empty-file analysis found that RDKit caps open S-group polymer ends with hydrogen (`Sgroups_Copolymer 01` becomes `CCC(C)C` + `CCCC`). Both cases produce well-formed InChIs; only a structure comparison exposes them.
- **RDKit hangs while reading, not in InChI.** Some structures stall RDKit even when parsed from SMILES: one `bjnano_5_145/m5445236-i13` structure hangs in aromaticity perception. All RDKit work in this comparison therefore ran in capped worker processes; 5 BChemXtract structures were left out for every reader, 4 compared rows timed out and 1 raised.

Limits

- Agreement, not correctness: the reference is BChemXtract 1.4.1, and only the findings marked as checked against the drawing say who is right.
- Every structure is compared through its SMILES. BChemXtract's InChIKey is the more reliable of its two outputs where they disagree (see findings), so stereo verdicts against BChemXtract overstate its InChI-level errors.
- Pairing by similarity can pair a structure with the wrong partner; wrong-skeleton verdicts all come from similarity pairs and may include such mis-pairings. Pairs by InChIKey or skeleton are unambiguous.
- Stereo is compared through CIP labels computed by RDKit for both sides, so the comparison does not depend on how each reader writes stereo.
- The reference uses `resolveRGroups = false`; with it on, BChemXtract enumerates Markush members that the other readers do not produce, which would only enlarge the unpaired counts.

Method. Structures: BChemXtract 1.4.1 via `testset/toolkits/BCXDump.java` (`xtractUnique(doc, info, false)`, SMILES and InChIKey per substance); RDKit, Indigo and ChemScanner via `testset/toolkits/run-toolkit-performance.py --dump`; Open Babel via `testset/toolkits/obabel-dump.sh` (`obabel file.cdx -ocan`), with reaction SMILES split into their molecules during pairing. **Pairing**: `testset/toolkits/pair_structures.py` deduplicates each side per file on InChIKey and pairs greedily by identical InChIKey, then identical InChIKey block 1, then Morgan fingerprint (radius 2, 2048 bits) Tanimoto ≥ 0.5; all RDKit work runs in a capped subprocess (60 s per file, 30 s per structure, 4 GB). **Comparison**: `testset/toolkits/run_compare.py` calls `testset/compare_structures.py` unchanged on every pair, in capped chunks (120 s per 25 rows, then 30 s per row), recording raised exceptions as errors and stalls as timeouts. Percentages are of compared structures. **Revision 2026-10-16**: the consensus conflicts were rechecked against BChemXtract's own InChIKeys (1.4.1, `main` at `f1db7bf`, and PR #227). BChemXtract's SMILES/InChIKey agreement was measured on `main` over the `.cdx` corpus by parsing its SMILES with CDK and generating the InChIKey, and by regenerating the InChI from each substance's own atom container without coordinates. Reproduce: `pair_structures.py REF.tsv PRED.tsv pairs.tsv`, then `run_compare.py pairs.tsv OUT`. Machine: Intel Core i5-12500H, Linux Mint 22.3, Python 3.12, RDKit 2026.09.1.